

ResNet-101

AttFPN

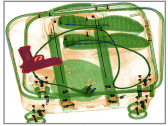
P_2–P_6

3 stages

RPN + cascade

boxes, classes

$b,\ c$



P_2–P_5

b

$b,\ c$

for each stage $s = 1, 2, 3$

P_2

RoIAlign

256 ch · 14 × 14

Mask head

separate per stage

$M^{(s)}$

C classes
28 × 28

Point refiner
Transformer or MLP

shared across stages
and refinement levels
56 × 56 → 112 × 112

$M^{(s,\,112)}$

C classes
112 × 112

σ

Mean over s

C classes
112 × 112

Select class
+ paste

$H \times W$

Threshold

$p \geq 0.5$

learned module

parameter-free operation

tensor

inference workflow